6. Low operating cost
7. More life

b. Applications

1. The compact size, longer life, low running and maintenance cost, instant glow makes these lamps suitable for all places where uniform illumination is required.
2. It is used in offices, shops, hotels, hospitals, cinema halls, residential buildings etc.

2.8 Neon and Halogen lamp

2.8.1 Neon lamp



Fig. 2.11 **Neon lamp**

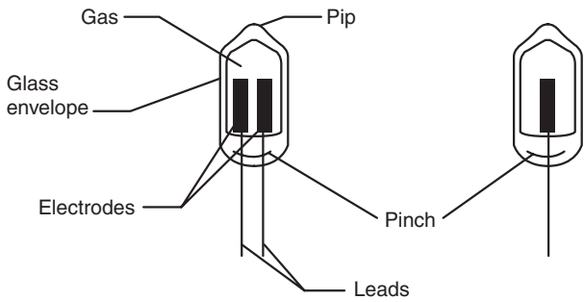


Fig. 2.12 **Line diagram of neon lamp**

a. Construction

The construction of neon lamp is shown in figure 2.11. A neon lamp is a small gas discharge lamp. The lamp commonly consists of a small glass capsule that contains a mixture of neon and different gases at a low pressure, and two terminals (an anode and a cathode). When adequate voltage is applied and sufficient current is supplied between the electrodes, the lamp produces an orange glow discharge. The glowing portion in the lamp is a thin region which is near to the cathode.

b. Applications

1. Neon lamps are generally used for advertising.
2. It is used as an indicator lamps
3. It is used for night lamps

2.8.2 Halogen lamp

Halogen lamp is a special type of tungsten filament lamp which was developed in 1959. In this, a small amount of halogen vapour is added to the inert gas of the bulb. The bulb is made of glass small in size. It operates at a temperature of 3000°C

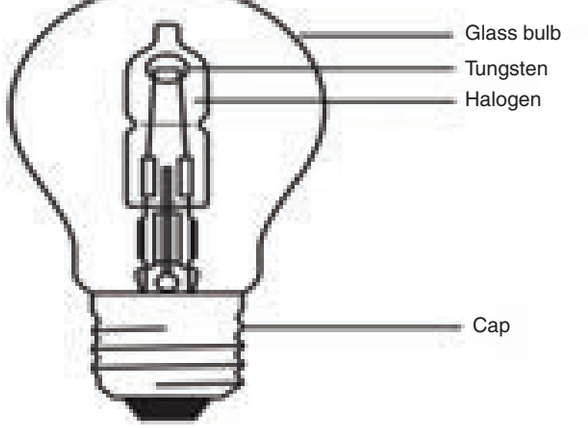


Fig. 2.13 **Halogen lamp**

Working principle

Halogen lamp is shown in figure 2.13. When the supply is given to the lamp, a filament glows, producing light. The halogen, an inert gas, causes the evaporated tungsten to settle back on the filament during cooling. That's why lamps

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